Exact stationary majority fraction in group-flip consensus dynamics on quenched hypergraph balls

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Abstract

Consensus dynamics on higher-order structures are usually analysed on annealed hypergraphs, where geometry enters only through degree or hyperedge-size distributions. We study two-colour group-flip dynamics on the quenched metric balls of a fixed hypergraph: when a node fires, every node of its colour inside its ball changes colour at once. The dynamics sees the hypergraph only through its 2-section graph. If every ball is a strict majority, every residual is erased by a single fire, and all balls have size $\boldsymbol{\beta}$, we prove that the stationary majority fraction is $\boldsymbol{E[M] = (2-q)/(3-2q)}$ with $\boldsymbol{q = \beta/n}$, together with a rotation identity and a two-point law for the inter-rotation cycle. We distinguish a geometric containment observable $\boldsymbol{c_G}$ from the one-step extinction probability $\tilde{\boldsymbol{c}}$ and give closed forms for both on one-dimensional hypergrids, confirmed by exhaustive enumeration. At intermediate coverage, numerically exact stationary solutions for all 17 one-dimensional cases with $\boldsymbol{n \leq 33}$ lie below the band curve and above an exactly solved annealed analogue, which motivates the conjecture that the band is an upper bound there. Finally, an exploratory regression on simulations of four graph families, with one graph per family and size, asks whether $\tilde{\boldsymbol{c}}$ accounts for the family dependence of $\boldsymbol{M}$ at intermediate coverage. It does not: the coefficient of $\tilde{\boldsymbol{c}}$ is negative ($\hat{\gamma} = \mathbf{-0.0196}$, cluster-bootstrap 95% interval $[\mathbf{-0.0222, -0.0169}]$), and including $\tilde{\boldsymbol{c}}$ increases the separation between families. Because $\tilde{\boldsymbol{c}}$ varies almost only within the one-dimensional family, the test does not identify an alternative mechanism.

Keywords: Voter dynamics, Hypergraphs, Group interactions, Consensus, Interacting particle systems, Exact stationary solutions

1 Introduction

Consensus-forming interacting particle systems, such as voter and majority-rule dynamics, are a standard testing ground of nonequilibrium statistical physics [1–6]. In the voter model a node adopts the state of a randomly chosen neighbour; this dyadic update produces purely diffusive collective behaviour on finite graphs. Interactions that involve groups of nodes are naturally described by hypergraphs and simplicial complexes [7, 8], and group constraints introduce nonlinear drift terms that are absent from dyadic models [9–12].

A recent example is the group-driven voter model (GVM) of Kim *et al.* [13]. There, a randomly chosen node selects one of its hyperedges, makes a fixed number of observations of the states in that hyperedge, and adopts the observed state only if the observations are unanimous and differ from its own. On annealed uniform hypergraphs, where hyperedge membership is redrawn at every step, the consensus time grows logarithmically with system size, faster than in the voter model, and is non-monotonic in the number of observations. The mean-field treatment relies on the annealed ensemble: all nodes are equivalent and geometry enters only through the hyperedge size. It therefore cannot address the question asked here: when the interaction sets are quenched metric balls of a fixed geometry, which geometric quantities control the stationary state, and which control the differences between geometry families?

We study a related but distinct rule, which we call group-flip dynamics. Each node fires at rate 1. When node i fires, every node that carries i's colour inside the ball $B_i(r)$ adopts a common target colour chosen uniformly among the other colours. In contrast to the GVM, the update is collective (an entire same-colour subset of the ball moves) and involves no unanimity condition. Two difficulties follow. First, because the moved set is a ball-local colour class, the natural state variable is the minority *set* rather than a colour count, so the dynamics is a finite Markov chain on subsets rather than a birth–death chain. Second, without structural assumptions on the geometry the stationary distribution of this chain has no known closed form, and no annealed closure is available because the balls are quenched. Although we use hypergraph terminology throughout, the dynamics depends on the hypergraph only through its 2-section graph (Remark 1); the object of study is the quenched metric-ball structure, not a higher-order interaction mechanism beyond pairwise topology.

Two structural conditions organise the analysis. Condition R1 ($\beta_i > n/2$ for all i) requires every ball to be a strict majority of the system; condition R2 ($\mathrm{diam}_H(V \setminus B_i) \leq r$ for all i) requires every residual, the complement of a ball, to be erasable by a single fire of a minority concentrated on it. We call the joint regime R1∧R2 the high-coverage regime. Whether R1 and R2 hold depends on the metric structure of the residuals, not only on ball sizes.

The contributions are of four kinds, which we keep separate throughout.

1. *Exact results (theorems).* In the high-coverage regime with equal ball sizes, the stationary majority fraction is $E[M] = (2-q)/(3-2q)$ with $q = \beta/n$ (Theorem 2). The derivation uses an epoch invariant on post-rotation minorities, an exact affine identity for the conditional mean minority mass, and a renewal–reward argument

over stationary cycles (Sec. 3). We also prove a rotation identity and a two-point law for the cycle length.
2. *Exact containment theory.* We distinguish the geometric containment c_G from the conditional containment $\tilde{c}$, which is the one-step extinction probability of a newly created minority, and derive closed forms and a three-region classification on one-dimensional hypergrids (Sec. 4).
3. *Exact finite-size computations and a conjecture.* Below the R2 threshold we solve the two-colour chain for six odd $n \leq 25$, with the stationary values certified as exact rationals, and extend the computation to every Region-II case with $n \leq 33$, odd and even (Sec. 5). All values lie below the band curve; we state the general inequality as Conjecture 1 and do not claim a proof. An annealed analogue with the same ball sizes, solved exactly, lies below both, which shows that the band is a property of the quenched balls.
4. *An exploratory statistical test.* We test, on simulations of four graph families, whether $\tilde{c}$ accounts for the family dependence of the stationary majority fraction at intermediate coverage (Sec. 6). The answer within this design is no; we report the result with its limitations and with sensitivity analyses for the resampling scheme and the family statistic.

Section 2 defines the model, the two structural conditions, and the containment observables. Section 3 derives the band. Section 4 develops the one-dimensional containment theory. Section 5 reports the exact finite-size solutions below the R2 threshold. Section 6 reports the exploratory test. Section 7 summarises numerical validation and limitations, Section 8 discusses open problems, and Section 9 concludes. The appendices collect the well-mixed limit, the cycle statistics, the newborn identity, and the containment algorithm. Complete proofs are given in the Supplementary Material (Online Resource 1, cited below as SM).

2 Model, definitions, and notation

2.1 Hypergraphs with metric interaction balls

Let $H = (V, E)$ be a hypergraph with $|V| = n$ nodes. The hypergraph distance $d(i, j)$ is the minimum number of hyperedge steps from i to j, where one step moves from a node to any node sharing a hyperedge with it. For $r \in \mathbb{N}$ the interaction ball of i is $B_i = \{j : d(i, j) \leq r\}$, of size $\beta_i = |B_i|$; when all balls have equal size we write β and define the coverage $q = \beta/n \in (0, 1]$. The balls are *quenched*: the sets B_i are fixed for the entire dynamics, in contrast with annealed ensembles in which hyperedge membership is redrawn at each step [13]. Since d is symmetric, $j \in B_i \iff i \in B_j$.

The complement $R_i = V \setminus B_i$ is the *residual* of i; it has $n - \beta_i$ nodes. A fire at i from a monochrome state creates exactly the minority R_i (Sec. 3). Two structural conditions organise the analysis:

R1: $\beta_i > n/2$ for all i (every ball is a strict majority);
R2: $\mathrm{diam}_H(R_i) \leq r$ for all i (every residual is contained in the ball of each of its own nodes).

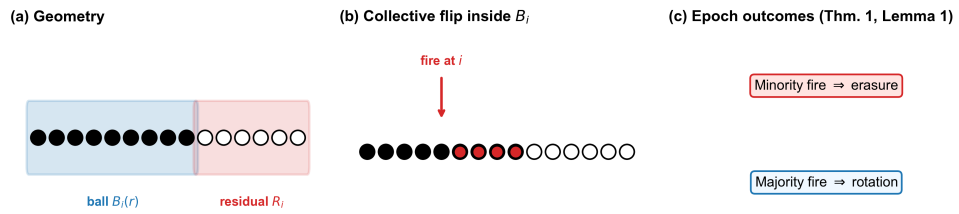


Fig. 1 Ball-constrained group-flip update. (a) 1D hypergrid with ball $B_i(r)$ (shaded) and residual R_i (open); hyperedges are sets of three consecutive nodes. (b) A fire at i: all nodes carrying σ_i inside B_i (filled) adopt a common target colour. (c) The two outcomes from a strict-majority state: a majority fire rotates the majority label (Theorem 1); a minority fire from a self-contained minority erases it (Lemma 1)

Under R1 a fire from the majority always changes the majority label (Theorem 1); under R2 a minority contained in a residual is erased by any fire of one of its own nodes, which is the statement $\tilde{c} = 1$ (Proposition 1).

On the one-dimensional hypergrid (a cycle whose hyperedges are the sets of three consecutive nodes) one hyperedge step moves at most two lattice sites, so $d(i,j) = \lceil \delta(i,j)/2 \rceil$, with δ the circular lattice distance, and

$$\beta = \min(n, 4r+1), \qquad \text{R1} \iff 2\beta > n, \qquad \text{R2} \iff \beta = n \text{ or } 3\beta \geq 2n-1. \quad (1)$$

When $\beta = 4r+1 < n$, R2 is equivalent to $n \leq 6r+2$ (SM Prop. B.2). Equation (1) is the finite-size form of the asymptotic statement that R1∧R2 holds for $q > 2/3$; we use the exact form throughout.

Remark 1 (Graph reduction) The dynamics depends on H only through its 2-section graph G, in which $i \sim j$ if i and j share a hyperedge: $B_i(r)$ is the graph-metric ball of radius r in G. For the 1D hypergrid, $G = C_n(\pm 1, \pm 2)$ and $B_i(r)$ is the closed neighbourhood of i in the r-th power $G^r = C_n(\pm 1, \ldots, \pm 2r)$. "Hypergraph" here therefore denotes a structured quenched neighbourhood rather than an interaction that is higher-order beyond its pairwise projection. We keep the hypergraph specification because the d-dimensional constructions (three-node hyperedges along each axis) are most naturally defined that way.

2.2 Group-flip dynamics on balls

Each node fires at rate 1, independently. When node i, of colour σ_i, fires: (i) the firing set is the set of nodes of colour σ_i inside B_i; (ii) a target colour is drawn uniformly among the $Q-1$ colours other than σ_i; (iii) every node of the firing set adopts the target colour simultaneously (Fig. 1). A single fire therefore moves an entire same-colour subset of a ball, and the outcome is deterministic given the target.

The well-mixed limit $B_i = V$ reduces the dynamics to a Markov chain on colour counts; its exact solution (pure death of live colours, consensus times, minority advantage, and three-colour winning probabilities) is summarised in Appendix A. For the

band and Region-II results we take $Q = 2$. The two-colour epoch chain used below is not claimed to describe $Q > 2$ trajectories, and extending the band to $Q > 2$ is outside the scope of this paper.

The primary observable is the stationary majority fraction M, the long-run time average of $\max(n_0, n_1)/n$, where n_c is the number of nodes of colour c; equivalently, its stationary expectation $E[M]$. We also use two cycle observables: the number K of fires between consecutive changes of the majority label (rotations), and the physical time T of the same cycle.

2.3 Containment observables

For residuals R_i define

$$c_G = \frac{1}{n^2} \sum_{i,j} \mathbf{1}\{R_i \subseteq B_j\}, \tag{2}$$

$$\tilde{c} = \frac{\sum_i \sum_{j \in R_i} \mathbf{1}\{R_i \subseteq B_j\}}{\sum_i |R_i|}, \tag{3}$$

with $\tilde{c} = 1$ if all residuals are empty. c_G is the probability that the ball of a uniformly chosen node contains a uniformly chosen residual; it is a purely geometric overlap statistic. $\tilde{c}$ is the probability that a node chosen uniformly inside the residual erases that residual in one fire. This is the conditioning realised by the dynamics when a newly created minority fires (Appendix C), so $\tilde{c}$ is the one-step extinction probability of a newborn minority. On vertex-transitive geometries with uniform balls c_G reduces to a single per-residual count (Proposition 2), whereas $\tilde{c}$ tracks whether residuals fit inside the balls of their own nodes and equals 1 in the high-coverage regime (Proposition 1).

2.4 Status labels and normalisation

Theorem, *Proposition*, *Lemma* and *Corollary* denote statements with complete proofs (main text or SM). *Conjecture* denotes a statement supported by numerical evidence only. *Protocol* denotes a measurement or statistical design; it is not a mathematical statement. All exact formulas use $q = \beta/n$.

3 Exact stationary band at high coverage

3.1 Rotation identity

Theorem 1 (Rotation identity) *Let $Q = 2$ and $\beta_i > n/2$ for every i (R1; equal balls are not assumed). From any strict-majority state, a rotation (change of the majority label) occurs if and only if the firing node belongs to the current majority, so the instantaneous rotation rate equals the majority size. Tie states are left at the first event and never re-entered; every stationary distribution assigns them zero mass.*

Proof sketch (full proof: SM Prop. C.1) Let $\mathcal{M}$ and $\mathcal{A}$ be the majority and minority classes and let the firing node be $i \in \mathcal{M}$. After the fire the class of the new colour is $\mathcal{A} \cup (\mathcal{M} \cap B_i)$

and the class of the old colour is $\mathcal{M} \setminus B_i$; the difference of their sizes is

$$|\mathcal{A}| + |\mathcal{M} \cap B_i| - |\mathcal{M} \setminus B_i| = (2\beta_i - n) + 2|\mathcal{A} \setminus B_i| > 0, \tag{4}$$

because $\beta_i > n/2$. The majority therefore changes label. If $i \in \mathcal{A}$, the minority can only shrink, so no rotation occurs. Each majority node fires at rate 1, so the rotation rate is $|\mathcal{M}|$. From a tie, Eq. (4) with $|\mathcal{M}| = |\mathcal{A}|$ shows that any fire produces a strict-majority state, and strict-majority states map to strict-majority states; hence ties are transient. Only the ball of the firing node enters, so equal balls are not needed. □

In the high-coverage regime the majority label is thus a fast variable that rotates at rate $|\mathcal{M}|$, while the minority set is the slow, geometrically constrained variable. The band curve follows from tracking the minority set between rotations.

3.2 Epoch chain and structural invariant

Let A_m be the minority set immediately after the m-th rotation, $a_m = |A_m|$, and i_m the firing node of that rotation.

Lemma 1 (Epoch invariant) *Under R1∧R2 with equal balls β, for every m: $A_m \subseteq R_{i_m}$. Consequently, by R2 applied to R_{i_m}, $A_m \subseteq B_j$ for every $j \in A_m$ (self-containment), and $a_m \leq n - \beta$.*

Proof sketch (full proof: SM Lemma C.2) From a monochrome state the first fire at i moves B_i to the new colour; since $\beta > n/2$, B_i is the new majority and the new minority is R_i. Given $A_m \subseteq R_{i_m}$, consider the next fire. If the firing node j lies in A_m, self-containment gives $A_m \subseteq B_j$, so the whole minority moves, the system becomes monochrome, and the following fire creates a minority $R_{i'}$. If the firing node i lies in the majority $\mathcal{M}_m$, only majority nodes inside B_i move (SM Remark C.1.1), and the new minority is $\mathcal{M}_m \setminus B_i \subseteq R_i$. Both branches preserve the invariant. □

Lemma 1 is where the quenched geometry enters: minorities observed just after a rotation are always contained in a residual, and self-containment is the property that makes the overlap computation below exact.

3.3 Exact conditional linearisation

Consider a majority fire at i, uniform in $\mathcal{M}$, from a minority A of size a. The new minority $\mathcal{M} \setminus B_i$ has size $n - \beta - a + |A \cap B_i|$.

Lemma 2 (Overlap identity) *Under the epoch invariant with equal balls β, for i uniform in $\mathcal{M}$,*

$$E\big[\, |A \cap B_i| \;\big|\; a \big] = \frac{a(\beta - a)}{n - a}. \tag{5}$$

Proof sketch (full proof: SM Lemma C.3) Exchange the order of summation in

$$\sum_{i\in\mathcal{M}} |A\cap B_i| = \sum_{i\in\mathcal{M}}\sum_{j\in A} \mathbf{1}\{j\in B_i\}$$

and use $j\in B_i \iff i\in B_j$: for fixed $j\in A$ the number of $i\in\mathcal{M}$ with $i\in B_j$ is $|B_j|-|B_j\cap A| = \beta - a$, by self-containment. Dividing by $|\mathcal{M}| = n-a$ gives the claim. □

A minority fire occurs with probability a/n; it erases the minority, and the following fire creates a new minority of size exactly $n-\beta$. A majority fire occurs with probability $1-a/n$, and by Lemma 2 the conditional mean of the new minority size is

$$\begin{aligned} E[a' \mid a,\ \text{majority fire}] &= n-\beta-a+E\big[|A\cap B_i|\,\big|\,a\big] \\ &= n-\beta-a+\frac{a(\beta-a)}{n-a} = \frac{(n-\beta)(n-2a)}{n-a}. \end{aligned} \tag{6}$$

Weighting the two branches gives

$$\begin{aligned} E[a_{m+1}\mid a_m=a] &= \frac{a}{n}(n-\beta)+\Big(1-\frac{a}{n}\Big)\frac{(n-\beta)(n-2a)}{n-a} \\ &= (1-q)(n-a). \end{aligned} \tag{7}$$

The conditional map is affine in a_m, so taking expectations against any distribution of a_m introduces no Jensen-type correction. This is the step on which the exact result rests.

3.4 Stationary mean and cycle structure

The epoch sets $\{A_m\}$ form a Markov chain on a finite family of subsets. Uniqueness of its stationary distribution is not needed, only existence. Let π be any invariant distribution and $\bar{x} = E_\pi[a]/n$. Taking π-expectations in Eq. (7) gives $\bar{x}n = (1-q)(n-\bar{x}n)$, hence

$$\bar{x} = \frac{1-q}{2-q}, \tag{8}$$

independently of π (SM Lemma C.5). Within a cycle between consecutive rotations, a second fire occurs if and only if the first fire is a minority fire, which has probability a_m/n given the epoch state. Averaging over π gives $P(K=2)=\bar{x}$ and $E[K]=1+\bar{x}$. Inter-event times are exponential with mean $1/n$ and independent of the colour process, so $E[T]=(1+\bar{x})/n$.

3.5 Renewal–reward argument and the band curve

During the first inter-event interval of a cycle the majority fraction is $1-a_m/n$; if a second interval occurs (probability a_m/n) the system is monochrome and the majority fraction is 1. The expected time-integrated majority fraction over one cycle is therefore $E\big[\int_{\text{cycle}} M\,dt\big] = \frac{1}{n}(1-\bar{x})+\frac{\bar{x}}{n} = \frac{1}{n}$. Cycles are stationary under π but not independent;

the ergodic ratio limit holds on each closed class of the epoch chain, and its value is the same on every class by Eq. (8) (SM Thm. C.6). Hence

$$E[M] = \frac{1/n}{(1+\bar{x})/n} = \frac{1}{1+\bar{x}}. \tag{9}$$

Theorem 2 (Exact band curve) *Assume $Q = 2$, R1, R2, and equal balls β. With $q = \beta/n$,*

$$E[M] = \frac{2-q}{3-2q}. \tag{10}$$

The function $f(q) = (2-q)/(3-2q)$ is defined for all $q \in (0, 1]$, with $f(1/2) = 3/4$, $f(2/3) = 4/5$ and $f(1) = 1$. Theorem 2 applies only on the R1∧R2 domain (exactly $3\beta \geq 2n - 1$ in 1D; asymptotically $q \geq 2/3$), where f takes values in $[(4n+1)/(5n+2), 1]$, which tends to $[4/5, 1]$ as $n \to \infty$ (for example $f = 11/14 \approx 0.786$ at the boundary case $n = 8$, $r = 1$, $\beta = 5$ of the 1D hypergrid). Values of f below the exact boundary $3\beta = 2n - 1$ (asymptotically $q = 2/3$) are evaluations of the closed form outside its domain of validity, not properties of the dynamics; we use them only as a comparator curve in Region II, where they are labelled as such. Equation (10) is exact for every finite n in the regime. Because R1 requires $\beta_i > n/2$, the radius must grow with n at fixed q (on d-dimensional hypergrids $r = \Theta(n^{1/d})$, SM Prop. B.3). The interaction sets therefore span a finite fraction of the system, and the large-n limit at fixed q is not a limit of short-range interactions; Theorem 2 is a finite-n statement about system-spanning quenched neighbourhoods.

Remark 2 (Where equal balls enter) Theorem 1 and the invariant $A_m \subseteq R_{i_m}$ of Lemma 1 hold without equal balls (SM Lemma C.2). Equal balls enter only through Lemma 2. For unequal balls the same exchange of summation gives $\sum_{i \in \mathcal{M}} |A \cap B_i| = \sum_{j \in A} (\beta_j - a)$, which depends on A through $\sum_{j \in A} \beta_j$, and a minority fire followed by a fire at i' creates a minority of size $n - \beta_{i'}$. The conditional mean of a_{m+1} is then not a function of a_m alone, and the closure of Eq. (7) is lost.

3.6 Threshold location and consistency conditions

Corollary 1 (Threshold location) *f is strictly increasing, with $f'(q) = (3-2q)^{-2}$, $f(1/2) = 3/4$ and $f(2/3) = 4/5$. Hence a crossing of $M = 3/4$ predicted by the closed form would lie at $q = 1/2$, outside the domain of the exact regime; the statement concerns the closed form, not the dynamics outside R1∧R2.*

Proposition 1 (R2 forces one-step extinction) *Under R2, $\tilde{c} = 1$. If balls are uniform, the 1D geometric containment is*

$$c_G = \min\left(1, \frac{\max(0, 2\beta - n + 1)}{n}\right), \tag{11}$$

where the cap at 1 is active only at full coverage $\beta = n$. Any theory of minority extinction must therefore reproduce $\tilde{c} = 1$ where the band holds (SM Prop. D.3).

Proposition 2 (c_G is a per-residual count) *On any vertex-transitive hypergraph with uniform balls, $c_G = n^{-1}|\{j : R_i \subseteq B_j\}|$, independent of i; on configuration models, $c_G = E_i\big[|\{j : R_i \subseteq B_j\}|/n\big]$ (SM Prop. D.1).*

Proposition 2 implies that on vertex-transitive geometries c_G is a function of the common ball and residual shape and carries no node-level information. It is not, however, a function of β alone: it can differ between families at fixed q, and all values must be computed by enumeration. The point is that c_G does not condition on the firing node lying in the residual, whereas the dynamics does; $\tilde{c}$ is the observable with that conditioning. Whether $\tilde{c}$ also accounts for effects outside the high-coverage regime is the empirical question of Sec. 6.

4 Exact containment on one-dimensional hypergrids

On the 1D hypergrid, B_i is an arc of β nodes and R_i an arc of $n - \beta$ nodes. When $\beta = 4r+1 < n$, β is odd, so each ball arc has a unique centre, and counting balls that contain a fixed residual arc is equivalent to counting admissible centres.

Proposition 3 (1D containment) *On the 1D hypergrid with $\beta = \min(n, 4r+1)$,*

$$c_G = \min\Big(1, \frac{\max(0, 2\beta - n + 1)}{n}\Big), \tag{12}$$

$$\tilde{c} = \begin{cases} 0, & 2\beta < n, \\ \dfrac{2\beta - n + 1}{n - \beta}, & 2\beta \geq n \text{ and } 3\beta < 2n - 1, \\ 1, & 3\beta \geq 2n - 1, \end{cases} \tag{13}$$

and $\tilde{c}$ is continuous at $3\beta = 2n - 1$.

Proof sketch (full proof: SM Prop. E.2) Let $\beta < n$. A ball arc of length β contains the residual arc of length $n - \beta$ if and only if its centre lies in an interval of $2\beta - n + 1$ positions, when this number is positive; dividing by the n centres gives c_G. At full coverage $\beta = n$ (possible whenever $4r + 1 \geq n$) residuals are empty; Eq. (12) gives $c_G = 1$ by the cap and Eq. (13) gives $\tilde{c} = 1$ by the empty-residual convention. For $\tilde{c}$: if $2\beta < n$, no ball contains R. If $2\beta = n$, then R is itself a ball of size β, and $R \subseteq B_j$ with $|B_j| = |R|$ forces $B_j = R$, whose unique centre lies in R; exactly one firing node is admissible. If $2\beta > n$ and $3\beta < 2n - 1$, the admissible centres form an interval of length $2\beta - n + 1$ strictly inside R. If $3\beta \geq 2n - 1$, the interval covers R and $\tilde{c} = 1$. □

The boundary case $2\beta = n$ is instructive: the residual is itself a ball, so $c_G = 1/n$ and $\tilde{c} = 1/\beta$, both non-zero; exactly one firing node, the centre of the residual arc, erases it in one fire. For $2\beta < n$ no ball contains a residual and both quantities vanish.

Proposition 4 (Three dynamical regions, 1D) *The 1D dynamics splits into Region I ($3\beta \geq 2n-1$: R1 and R2 hold, the band applies, $\tilde{c} = 1$); Region II ($n/2 < \beta < (2n-1)/3$: R1 holds,*

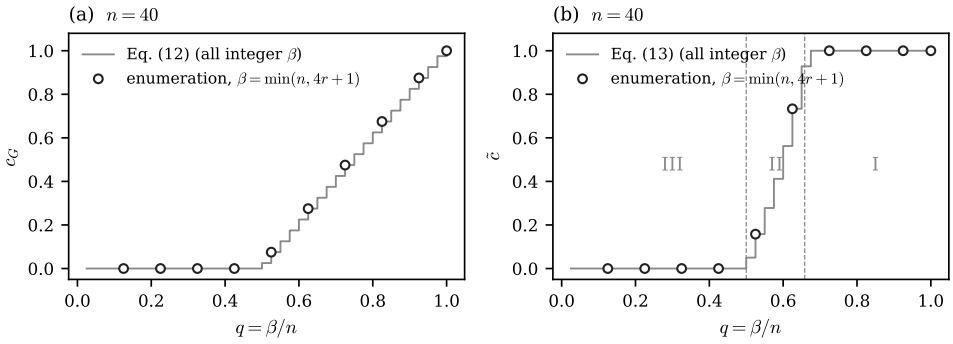


Fig. 2 Exact 1D containment for $n = 40$. (a) c_G and (b) $\tilde{c}$ against q: closed forms Eqs. (12)–(13) evaluated at every integer β (grey steps) and values obtained by exhaustive enumeration of all ball–residual pairs for the realised ball sizes $\beta = \min(n, 4r + 1)$ (open circles). Dashed lines in (b) mark the region boundaries $\beta = n/2$ and $\beta = (2n - 1)/3$. The boundary case $2\beta = n$ requires even n and is checked separately at $(18, 2)$. Data: `results/validation/containment_bruteforce.json`

R2 fails, $0 < \tilde{c} < 1$*); the boundary case* $2\beta = n$ *(*$c_G = 1/n$*,* $\tilde{c} = 1/\beta$*); and Region III (*$2\beta < n$*: R1 fails,* $\tilde{c} = 0$*). Asymptotically in* q *these are* $q \geq 2/3$*,* $1/2 < q < 2/3$*, and* $q \leq 1/2$*.*

In Region II residuals are not erasable in one fire, so the fate of a minority is a multi-step geometric problem. Equations (12) and (13) agree exactly, in rational arithmetic, with exhaustive enumeration of all ball–residual pairs in all 192 cases (n, r) with $12 \leq n \leq 40$, including the boundary case $(n, r) = (18, 2)$, where $c_G = 1/18$ and $\tilde{c} = 1/9$ (Fig. 2; SM Sec. E).

5 Exact finite-size stationary solutions below the R2 threshold

5.1 Exact two-colour chain

For odd n, the two-colour dynamics can be described by the minority set A, $|A| \leq (n - 1)/2$, which evolves as a finite Markov chain. Of the 2^{n-1} possible minority sets, far fewer are reachable from the monochrome configuration: 73, 481, 2177, 2689, 11777 and 51201 for $n = 9, 15, 17, 21, 23, 25$. In every case the reachable set is a single irreducible class. The dynamics is equivariant under the dihedral group D_n and under exchange of the two colours, so the chain lumps exactly onto orbits; the $n = 25$ case reduces to 1088 orbits. We computed the stationary distribution in two independent ways (SM Protocol F.3): in floating point on the full reachable set (maximal L^1 residual 4.1×10^{-14}), and in exact rational arithmetic on the lumped chain, with the solution verified by exact substitution. Tie states are transient, so the chain is also well defined for even n; the exact rationals below are for odd n, and Sec. 5.3 adds even n.

Table 1 Stationary majority fraction in Region II of the 1D hypergrid, for all six odd $n \leq 25$ with $n/2 < \beta < (2n-1)/3$ and $\beta = 4r+1$. The exact value is a rational number obtained by exact arithmetic on the symmetry-lumped chain and agrees with an independent floating-point solution of the unreduced chain to $\leq 1.3 \times 10^{-14}$. $f(q)$ is the band function, Eq. (10), evaluated outside its domain of validity as a comparator. All differences are negative.

n	r	β	q	$\tilde{c}$	$E[M]$ (exact)	decimal	$f(q)$	$E[M] - f(q)$
9	1	5	0.5556	1/2	43/57	0.7543860	0.7647059	−0.0103199
15	2	9	0.6000	2/3	65/84	0.7738095	0.7777778	−0.0039683
17	2	9	0.5294	1/4	149/201	0.7412935	0.7575758	−0.0162822
21	3	13	0.6190	3/4	265/339	0.7817109	0.7837838	−0.0020729
23	3	13	0.5652	2/5	72/95	0.7578947	0.7674419	−0.0095471
25	3	13	0.5200	1/6	319/433	0.7367206	0.7551020	−0.0183815

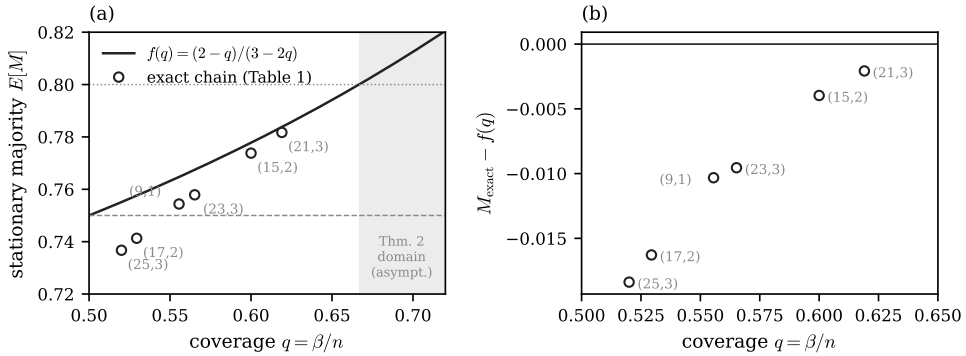


Fig. 3 Exact Region-II values and the band function. (a) Exact stationary majority fraction for the six cases of Table 1 (open circles, labelled (n, r)) and $f(q) = (2-q)/(3-2q)$ (solid). The shaded area marks $q \geq 2/3$, the asymptotic location of R1∧R2, where Theorem 2 applies; to its left f is a comparator only. Horizontal guides at 3/4 and 4/5. (b) Differences $E[M] - f(q)$; all are negative (Conjecture 1)

5.2 Results and Conjecture 1

Conjecture 1 *In Region II, $E[M] \leq f(q) = (2-q)/(3-2q)$ for all n and all admissible r.*

In all six cases the exact stationary value lies strictly below the comparator curve, by between 0.0021 and 0.0184 (Table 1, Fig. 3). The rational values show no evident common form in (n, r), and six cases are not sufficient to establish the inequality in general; we therefore state it as a conjecture and record the exact values as reference data for a future theory of Region II.

Table 2 All Region-II cases of the 1D hypergrid with $n \le 33$ ($n/2 < \beta < (2n-1)/3$, $\beta = 4r+1$). "States": configurations reachable from the monochrome state, up to colour exchange. $E[M]$: stationary majority fraction of the quenched dynamics, computed by the independent solver of Sec. 5.3 (L^1 residual $< 5 \times 10^{-13}$). $f(q)$: band function used as a comparator. $E[M]_{\text{ann}}$: annealed analogue with the same β, interaction set redrawn at every fire

n	r	β	q	$\tilde{c}$	states	$E[M]$	$f(q)$	$E[M]-f(q)$	$E[M]_{\text{ann}}$
9	1	5	0.5556	1/2	73	0.754386	0.764706	−0.0103	0.733333
15	2	9	0.6000	2/3	481	0.773810	0.777778	−0.0040	0.750000
16	2	9	0.5625	3/7	1025	0.756831	0.766667	−0.0098	0.732639
17	2	9	0.5294	1/4	2177	0.741294	0.757576	−0.0163	0.717647
21	3	13	0.6190	3/4	2689	0.781711	0.783784	−0.0021	0.757576
22	3	13	0.5909	5/9	5633	0.769553	0.775000	−0.0054	0.744318
23	3	13	0.5652	2/5	11777	0.757895	0.767442	−0.0095	0.732441
24	3	13	0.5417	3/11	24577	0.746914	0.760870	−0.0140	0.721726
25	3	13	0.5200	1/6	51201	0.736721	0.755102	−0.0184	0.712000
27	4	17	0.6296	4/5	13825	0.785965	0.787234	−0.0013	0.761905
28	4	17	0.6071	7/11	28673	0.776560	0.780000	−0.0034	0.751190
29	4	17	0.5862	1/2	59393	0.767367	0.773585	−0.0062	0.741379
30	4	17	0.5667	5/13	122881	0.758488	0.767857	−0.0094	0.732353
31	4	17	0.5484	2/7	253953	0.750000	0.762712	−0.0127	0.724014
32	4	17	0.5312	1/5	524289	0.741958	0.758065	−0.0161	0.716283
33	4	17	0.5152	1/8	1081345	0.734396	0.753846	−0.0195	0.709091
33	5	21	0.6364	5/6	67585	0.788618	0.789474	−0.0009	0.764706

5.3 Extended computation and an annealed comparison

To test Conjecture 1 beyond the six rational cases we solved the stationary chain for every Region-II case of the 1D hypergrid with $n \le 33$, odd and even (17 cases; Table 2). The code is independent of the two solvers of Sec. 5: its states are all configurations reachable from the monochrome state, lumped only under colour exchange, and the stationary law is obtained by power iteration of the lazy jump chain, which is legitimate because the total jump rate equals n in every state. The L^1 residual is below 5×10^{-13} in every case. The code reproduces the six values of Table 1 to 10^{-12} and the reachable-set sizes quoted above. All 17 values lie below $f(q)$, by between 0.0009 and 0.0195.

Annealed comparison. To isolate the role of the quenched geometry we also solved an annealed analogue with the same ball sizes: at every fire of i the interaction set is redrawn as i together with $\beta-1$ nodes chosen uniformly from $V \setminus \{i\}$. The dynamics then reduces to a Markov chain on the number of nodes of one colour, whose stationary law we compute by a direct linear solve. In every case of Table 2 the annealed value lies below the quenched one, by between 0.021 and 0.026. At high coverage the annealed value also lies strictly below the band: at $n = 400$ it is 0.7949, 0.8215, 0.8501 and 0.9154 at $q = 0.70$, 0.75, 0.80 and 0.90, against $f(q) = 0.8125$, 0.8333, 0.8571 and 0.9167. The band is therefore a property of quenched balls. Under R2 a minority fire erases the whole self-contained minority (Lemma 1); a freshly drawn set removes only the part of the minority that it happens to contain.

5.4 Interpretation

The sign of the deviation is the informative part. Outside R2 a residual is not erasable in one fire: with probability $1-\tilde{c}$ the first fire of a newborn minority from inside it leaves part of the minority in place, possibly as several pieces. The band calculation

assumes one-fire erasure, and longer-lived minorities would lower the time-averaged majority fraction. The negative deviations are consistent with this picture. We have not, however, isolated it: the exact chain gives the stationary mean, not a decomposition by mechanism, and no theorem here identifies multi-step survival as the cause of the deviation.

Remark 3 (Cycle form of Conjecture 1) Under R1 alone, every majority fire is a rotation and no minority fire is (SM Prop. C.1). Firing epochs form a Poisson process of rate n independent of the configuration, so the state seen at firing epochs has the stationary law; this is a PASTA argument applied to the firing epochs, not to the rotation epochs, which are a state-dependent thinning. The long-run fraction of fires that are rotations is therefore $E[M]$, and $E[M] = 1/E[K]$, with both sides understood class-wise if the chain is reducible. Conjecture 1 is thus equivalent to $E[K] \geq (3-2q)/(2-q)$ in Region II. Across all 17 cases of Table 2, odd and even, the deviation increases monotonically with q (from -0.0195 at $q = 0.515$ to -0.0009 at $q = 0.636$), whereas $\tilde{c}$ does not (for example, it is $1/2$ at $q = 0.556$ and $2/5$ at $q = 0.565$). Within this set, q organises the deviation better than $\tilde{c}$.

6 Exploratory test of the one-step containment hypothesis

6.1 Hypothesis and design

At intermediate coverage, simulations of four geometry families show family-dependent stationary majority fractions at the same (n, q) (Fig. 4a). A natural explanation is that $\tilde{c}$, the one-step extinction probability of a newborn minority, mediates this dependence: families with larger $\tilde{c}$ would show larger M, and controlling for $\tilde{c}$ would reduce the family effects.

Protocol 1 (Exploratory regression test) *Families and design.* 1D cycle; 2D torus with king (8-neighbour) adjacency; configuration model with degree sequence $k = 6$, projected to a simple graph; Watts–Strogatz (WS) with $k = 6$, $p = 0.1$. Sizes $n \in \{100, 144, 400\}$; coverages $q \in \{0.55, 0.60, \ldots, 0.80\}$; one graph realisation per (family, n), drawn with a fixed seed; three runs (seeds) per (family, n, q): 216 runs in total.

Interaction sets. To remove ball-size heterogeneity, each B_i is truncated to $\beta = \mathrm{round}(qn)$ nodes by breadth-first layering from i, with ties in the last layer broken at random using the run seed. Consequently the three runs of a cell differ in the last-layer tie breaking as well as in the dynamics; $\tilde{c}$ differs between runs of the same cell in 15 of the 72 cells. Truncated sets need not satisfy $j \in B_i \Leftrightarrow i \in B_j$, so this is a modified interaction-set model, not the native metric-ball model of Theorem 2.

Measurement. $Q = 2$; 2×10^5 fires per run, the first 5×10^4 discarded; M and its Monte Carlo standard error (SE) from 75 batch means of 2000 fires. c_G and $\tilde{c}$ by exact enumeration on the truncated sets.

Model. Weighted least squares on the 216 runs, weights $1/\mathrm{SE}^2$:

$$M \sim q + C(\text{family}) + C(n) \quad \text{(reduced)}, \qquad M \sim q + \tilde{c} + C(\text{family}) + C(n) \quad \text{(full)},$$

with γ the coefficient of $\tilde{c}$. This is an exploratory regression controlling for coverage (linearly), system size and family fixed effects; it is not a matched or full (n, q)-cell fixed-effects design.

Table 3 Coefficient γ of $\tilde{c}$ and the family-norm reduction (six pairwise contrasts) for the primary analysis and sensitivity variants. Point estimates are full-sample fits; intervals are bootstrap percentile 95% intervals. Negative reductions mean that adding $\tilde{c}$ increases the separation between families. *Primary analysis (Protocol 1). All rows use the same 216 runs and $B = 2000$ replicates. "Unstratified cluster": two-level cluster bootstrap without family or coverage stratification; "within cells": runs resampled only within (family, n, q) cells, i.e. Monte Carlo noise alone. Further variants in SM Table S4. Abbreviations: primary, the specification of Protocol 1; lin./categ. q, coverage entered linearly/categorically; FE, fixed effects.

Specification	Resampling	$\hat{\gamma}$	95% interval	Reduction	95% interval
primary: lin. q, $C(n)$	stratified cluster*	-0.0196	$[-0.0222, -0.0169]$	-0.98	$[-1.129, -0.849]$
primary: lin. q, $C(n)$	unstratified cluster	-0.0196	$[-0.0236, +0.0022]$	-0.98	$[-1.365, -0.001]$
primary: lin. q, $C(n)$	runs iid	-0.0196	$[-0.0223, -0.0170]$	-0.98	$[-1.249, -0.753]$
primary: lin. q, $C(n)$	within cells	-0.0196	$[-0.0200, -0.0192]$	-0.98	$[-1.021, -0.937]$
lin. q, no $C(n)$	stratified cluster	-0.0191	$[-0.0221, -0.0165]$	-0.95	$[-1.116, -0.821]$
categ. q, $C(n)$	stratified cluster	-0.0185	$[-0.0208, -0.0161]$	-0.92	$[-1.074, -0.802]$
(n, q)-cell FE	stratified cluster	-0.0186	$[-0.0215, -0.0159]$	-0.93	$[-1.119, -0.772]$

Family-effect norm. With the four family effects $(\alpha_{1D}, \alpha_{2D}, \alpha_{WS}, \alpha_{config})$, $N = \left[\sum_{f<g}(\alpha_f - \alpha_g)^2\right]^{1/2}$ over all six pairs; N does not depend on the reference family. The reduction is $(N_{reduced} - N_{full})/N_{reduced}$; negative values mean that adding $\tilde{c}$ increases the separation.

Inference. Cluster bootstrap that respects the hierarchy: within each family, the three (family, n) clusters are resampled with replacement; within each selected cluster, the three runs are resampled within each of the six coverage strata. $B = 2000$, percentile 95% intervals.

Acceptance criteria (fixed before the data were analysed). (a) γ positive with interval excluding zero; (b) reduction $\geq 30\%$.

Analysis record. Only the acceptance criteria were fixed before the data were analysed. The pre-specified analysis resampled runs independently and measured family separation relative to a single reference family. The resampling scheme and the family statistic of Protocol 1 were adopted afterwards, because two simpler choices are inadequate for this design: a statistic built from the three contrasts among the non-1D families is neither reference-invariant nor sensitive to the 1D family, the only one in which $\tilde{c}$ varies appreciably; and a two-level cluster bootstrap without stratification omits at least one whole family in 61 of 500 replicates and does not preserve the coverage strata. Both are reported as sensitivity analyses (Table 3; SM Note F.5). Because the resampling scheme and the statistic were chosen after the data had been analysed, the test is exploratory.

6.2 Result

Neither acceptance criterion is met (Table 3). The full-sample estimate is $\hat{\gamma} = -0.0196$, with 95% interval $[-0.0222, -0.0169]$: the coefficient is negative, opposite in sign to the hypothesis, and its interval excludes zero. Adding $\tilde{c}$ to the model does not reduce the separation between families; it increases it (reduction -0.98, interval $[-1.13, -0.85]$; fraction of replicates reaching the 30% threshold: 0.000). Both conclusions are unchanged, with intervals excluding zero, when coverage enters as a

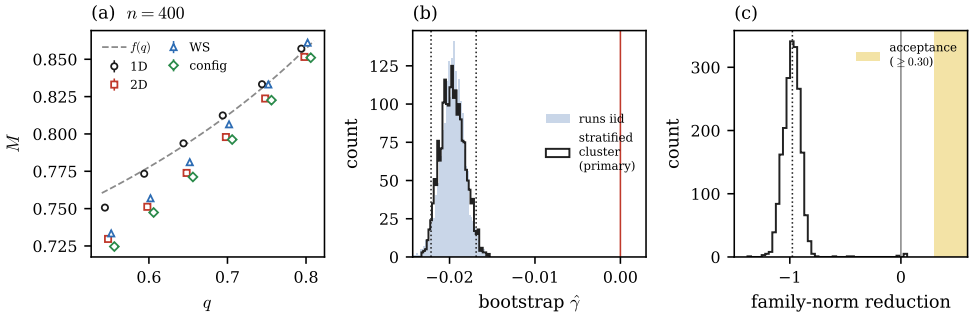


Fig. 4 Exploratory test of the one-step containment hypothesis. (a) Simulated M against q at $n = 400$ for the four families: mean of the three runs per cell with the standard error of that mean (smaller than the markers); horizontally offset for legibility. The band function $f(q)$ is shown as a comparator; the truncated interaction sets do not satisfy the assumptions of Theorem 2. (b) Bootstrap distributions of $\hat{\gamma}$: family-stratified cluster bootstrap (primary, outline, dotted 95% limits) and independent resampling of runs (filled, sensitivity). (c) Bootstrap distribution of the family-norm reduction under the primary scheme; the acceptance region (reduction ≥ 0.30) is shaded. Data: `data/processed/measurements_v2.csv`; `results/tables/reanalysis/`

categorical variable, with full (n, q)-cell fixed effects, and without the size term (Table 3). The point estimates are similar without weights ($\hat{\gamma} = -0.0181$) and when the 72 cell means are fitted instead of the 216 runs ($\hat{\gamma} = -0.0188$; SM Table S4). Leaving out any one of the 12 clusters gives $\hat{\gamma}$ between -0.0208 and -0.0186.

The increase of the family separation has a simple source. The 1D family has the largest $\tilde{c}$ and the largest M. With $\hat{\gamma} < 0$, the term $\hat{\gamma}\tilde{c}$ lowers the fitted value of the 1D cells, and the 1D fixed effect rises to compensate, moving it further from the other three families. The three-contrast statistic omits the 1D family and therefore registers no change (reduction -0.047, interval $[-0.167, +0.018]$).

6.3 Interpretation and caveats

The negative result is consistent with the exact theory where it applies ($\tilde{c} = 1$ throughout Region I, Proposition 1) and fails to support the mediation hypothesis outside it. Four caveats limit what it shows.

First, identification. Once family fixed effects are included, γ is identified by variation of $\tilde{c}$ within families. That variation is almost entirely in the 1D family (96.6% of the within-family sum of squares); $\tilde{c}$ is identically zero in all 2D cells and at most 0.241 (WS) and 0.036 (configuration). Within 1D, $\tilde{c}$ rises steeply between $q = 0.55$ and 0.65 and is then constant at 1. Over the same range the advantage of the 1D family over the others shrinks (Fig. 4a). The negative coefficient records this co-variation; even with full (n, q)-cell fixed effects it cannot separate an effect of one-step containment from any other property of the 1D family that changes as it enters Region I. We therefore read the result as evidence against the mediation hypothesis in this design, not as evidence that one-step containment lowers M.

Second, inference. Only 12 (family, n) clusters exist, three per family, and each is a single graph realisation. The intervals describe uncertainty about this realisation set;

they are not graph-ensemble intervals, and the family offsets are properties of these specific graphs. The primary interval is about as wide as the one from resampling runs independently, and about six times wider than resampling runs only within cells, which reflects Monte Carlo noise alone (Table 3). The unstratified two-level scheme gives a much wider interval that includes zero; its positive values come only from replicates in which at least one family is absent (22% of those replicates; none of the complete ones). With three clusters per family the bootstrap may still understate between-realisation variability.

Third, the truncated interaction sets differ from native metric balls, so a negative result here does not test a native-ball mechanism. In Region I, native-ball simulations reproduce the band to Monte Carlo precision (Sec. 7), and the 1D truncated sets, which are contiguous arcs, give M within 2×10^{-4} of $f(\beta/n)$ at $q \geq 0.70$. The other three families lie below f throughout; we do not attempt to explain this offset. The coverage regressor is the nominal grid value q rather than β/n (they differ by at most $0.5/n$); using β/n gives $\hat{\gamma} = -0.0201$.

Fourth, a null result about $\tilde{c}$ is not evidence for a particular alternative. The exact Region-II values (Sec. 5) are consistent with multi-step minority survival, but neither analysis isolates that mechanism.

A stronger test would need independent variation of $\tilde{c}$ at fixed (n, q) across families, for example through a one-parameter family of geometries that interpolates between the 1D hypergrid and a random graph.

7 Validation status and limitations

Static checks. Equations (12)–(13) agree exactly with enumeration in 192 cases (Sec. 4). The six Region-II stationary values were computed twice, by independent codes, and certified as exact rationals (Sec. 5); a third, independent code reproduces them and extends the computation to all Region-II cases with $n \leq 33$ (Sec. 5.3).

Simulation checks. An independent continuous-time simulator (10 replicas of 2×10^6 fires per case; SM Protocol F.3) was compared with the exact results. On native 1D balls in Region II, all six cases of Table 1 are reproduced within 4.6×10^{-5} (largest $|z| = 1.6$). In Region I, on native 1D balls ($n = 400, 401$) and 2D hypergrids ($L = 12, 30$) for which R1 and R2 were verified by breadth-first search, M agrees with Eq. (10) within 2.1×10^{-5} (largest $|z| = 0.8$), and the cycle statistics agree with Appendix B: largest $|z| = 1.9$ for $E[K]$ and 1.9 for the squared coefficient of variation CV_T^2 of T. The newborn identity (Appendix C) agrees with the enumerated $\tilde{c}$ within binomial error, both on native 1D balls (e.g. $n = 15$, $r = 2$: 29 489 trials, 0.6664 ± 0.0027 against $\tilde{c} = 2/3$) and on the truncated sets of Sec. 6 (all 18 cells with $0 < \tilde{c} < 1$, largest $|z| = 1.85$; SM Remark F.1.1).

Limitations. (i) $Q = 2$ throughout the band and Region-II results. (ii) The band requires equal ball sizes and R1∧R2. (iii) Region II is solved in exact rational arithmetic only for six small odd n, and in floating point for all cases with $n \leq 33$; Conjecture 1 is unproven. (iv) The test of Sec. 6 uses four families, three sizes, one graph realisation per (family, n), and truncated rather than native interaction sets; coverage enters linearly in the primary model. (v) No mechanism for the family dependence of M is identified.

8 Discussion and open problems

The exact band holds because, under R1∧R2 with equal balls, the conditional mean of the post-rotation minority mass is affine in the current mass. Outside R2 this structure is lost, and the six exact Region-II solutions show that the stationary majority fraction then lies below the closed form. A theory of Region II would need to account for minorities that survive their first fire; a natural candidate is a multi-step containment functional, such as the probability of extinction within k fires or the survival statistics of residual fragments. The exploratory test shows that the one-step quantity $\tilde{c}$, within its limited power, does not account for the family dependence observed at intermediate coverage. The gap between the configuration-model and 2D families also remains open (SM Open problem G.4).

Relation to annealed theory. The results complement the mean-field analysis of Kim *et al.* [13] for a different update rule. Where annealed theory yields leading-order prefactors of the logarithmic exit time, the quenched ball structure yields an exact stationary value (Theorem 2), because the epoch chain is finite and the conditional map, Eq. (7), is affine. In the GVM the nonlinearity comes from the number of unanimous observations; here it comes from the collective move of a ball-local colour class, so residual geometry rather than sampling is the controlling variable. The optimum in exit time found in Ref. [13], which arises from a competition between diffusive small groups and slow drift in large groups, has no direct counterpart here. Within the present model, the annealed analogue of Sec. 5.3 lies below the quenched values both in Region II and at high coverage, so the band is a consequence of quenching rather than of the ball size alone.

9 Conclusion

Group-flip consensus on quenched hypergraph balls admits an exact stationary majority fraction $E[M] = (2-q)/(3-2q)$ at high coverage, together with a rotation identity and a two-point cycle law. An exact containment theory separates the geometric observable c_G from the one-step extinction probability $\tilde{c}$, with closed forms and a three-region classification in one dimension. Below the R2 threshold, exact solutions for all 17 one-dimensional cases with $n \leq 33$ lie below the band curve and above an annealed analogue with the same ball sizes, which motivates Conjecture 1. An exploratory test finds no evidence that $\tilde{c}$ accounts for the family dependence of the stationary state at intermediate coverage; adding it to the model increases the family separation. Explaining Region II, and the family dependence, will require going beyond one-step containment.

Supplementary information. Online Resource 1 (Supplementary Material, cited in the text as SM) contains the complete proofs, the protocols, the validation records and the numerical tables referred to in the text.

Declarations

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Competing interests. The author has no relevant financial or non-financial interests to disclose.

Ethics approval and consent to participate. Not applicable.

Consent for publication. Not applicable.

Data availability. The data are archived at Zenodo [14] (https://doi.org/10.5281/zenodo.24014101). Every number in Secs. 5–7 is generated from the released outputs by `scripts/make_numbers_tex.py`, except those of Sec. 5.3, which are produced by `scripts/region2_exact.c`, `scripts/annealed_baseline.py` and `scripts/make_region2_table.py`.

Materials availability. Not applicable.

Code availability. The code is archived at Zenodo [14] (https://doi.org/10.5281/zenodo.24014101) and developed at https://github.com/phmldtngyn/Group-flip-consensus-on-hypergraphs.

Author contributions. Not applicable (single-author article).

Use of generative AI and AI-assisted technologies. During the preparation of this work the author used Claude Opus 5.5 and Claude Sonnet 5.5 (Anthropic) for code development, LaTeX editing, manuscript review and language editing, and Qwen 3.8 Max (Alibaba) for literature search. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Appendix A Well-mixed limit

For $B_i = V$, a fire at a node of colour k moves the whole class k to a target $l \neq k$ chosen uniformly among $Q-1$ colours. The count vector x therefore jumps by $x_k(e_l - e_k)$ at rate $x_k/(Q-1)$ for each target, and the count process is Markov on the simplex (SM Lemma A.1). The number L_t of live colours is a pure-death chain: a fire from a live class merges it into another live class with probability $(\ell-1)/(Q-1)$ when ℓ colours are live, and the total firing rate is n; hence $L \to L-1$ at rate $n(\ell-1)/(Q-1)$, with $L=1$ absorbing (SM Lemma A.2). Summing the mean holding times gives the first-consensus time

$$E[T_{\mathrm{cons}}] = \frac{Q-1}{n} \sum_{\ell=2}^{Q} \frac{1}{\ell-1} = \frac{Q-1}{n} H_{Q-1},$$

so $E[T_{\mathrm{cons}}] = 1/n$ for $Q=2$ (SM Thm. A.3). For $Q=2$ the first fire produces consensus, and from $(a, n-a)$ the class of size a wins with probability $1-a/n$: the smaller class is more likely to win (SM Thm. A.4). For $Q=3$ with state $(a,b,0)$ and $D = a^2+3ab+b^2$, the winning probabilities are $P_1 = b(a+b)/D$, $P_2 = a(a+b)/D$ and $P_3 = ab/D$ (SM Thm. A.5). For general Q the winning probabilities are the unique solution of a linear first-step system whose transient block is invertible (SM Thm. A.6).

Appendix B Cycle statistics

Under the assumptions of Theorem 2, every cycle between rotations contains $K \in \{1, 2\}$ fires, with $P(K = 2) = \bar{x} = (1 - q)/(2 - q)$ (Lemma 1 and Sec. 3). Hence

$$E[K] = 1 + \bar{x} = \frac{1}{E[M]}, \qquad E[T] = \frac{1 + \bar{x}}{n}, \qquad \mathrm{CV}_T^2 = \frac{7 - 8q + 2q^2}{(3 - 2q)^2},$$

where the last expression follows from the two-point mixture of exponential inter-event times (SM Thm. C.8). Measuring K and T in simulation tests the cycle decomposition independently of the band.

Appendix C Newborn identity

A minority is a *newborn* if it is created by the first fire after a monochrome state; for a fire at i the newborn is R_i. Conditional on the next event being a fire of a newborn node, the firing node is uniform in R_i and, with $Q = 2$, the target is the majority colour; the newborn is erased if and only if $R_i \subseteq B_j$ for the firing node j. The conditional erasure probability is therefore $|\{j \in R_i : R_i \subseteq B_j\}|/|R_i|$, and averaging over i with weights $|R_i|$ gives $\tilde{c}$, Eq. (3). This identity links the dynamic conditioning to the geometric definition: a correct simulator must reproduce $\tilde{c}$ within Monte Carlo error (SM Prop. F.1; Sec. 7).

Appendix D Containment by intersection counting

For uniform balls, $R_i \subseteq B_j$ is equivalent to $|B_i \cap B_j| = 2\beta - n$ (SM Prop. D.4). Hence c_G is computed exactly by an $O(n^2)$ count of the pairs (i, j) with $\mathrm{popcount}(B_i \,\&\, B_j) = 2\beta - n$, divided by n^2; if $2\beta < n$, $c_G = 0$. The enumeration for $\tilde{c}$ restricts j to R_i and normalises by $\sum_i |R_i|$ (SM Prop. F.2).

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